Interpretability-Driven Dermatological Diagnosis via Multimodal Foundation Models and Quantifiable Severity Index (QSI)

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***Abstract*—Skin cancer still stands as one of the most common forms of cancer globally, with melanoma being the deadliest form of skin cancer. The diagnosis of skin diseases requires multiple modalities, which depend on a combination of visual morphological characteristics with patient clinical data. In this paper, we present Dino-Derm, a new diagnostic pipeline that utilizes DINOv2, a cutting-edge self-supervised ViT architecture, as a feature extractor coupled with clinical data (age, sex, and anatomic location) via a late-fusion network. Our method aims to address two major challenges faced by current approaches, namely: (i) the use of image information only, devoid of any clinical context, and (ii) the lack of interpretability inherent in deep learning models. To overcome these two barriers, we propose the Quantifiable Severity Index (QSI), an interpretable index that quantitatively evaluates lesion severity by using attention rollout analysis and feature contrast. Evaluations based on the HAM10000 dataset [1] demonstrate the superiority of our approach compared to baselines in both accuracy (82%) and interpretability.**

***Index Terms*—Dermoscopy, DINOv2, Multimodal Fusion, Self-Supervised Learning, Explainable AI (XAI), Attention Rollout, Clinical Decision Support, Skin Lesion Classification.**

I. Introduction

Skin cancer is considered the most common cancer in the world, with about 9,500 people in the United States diagnosed every day [1]. Out of skin cancer types, melanoma is the deadliest, and its long-term survival is less than 10%, even though significant advances have been made in treating this disease recently. Early diagnosis using dermoscopic examination can be done only by experienced specialists, which makes it the key for successful treatment [13].

The development of Deep Learning models, especially Convolutional Neural Networks (CNN) and Vision Transformers (ViTs), enabled the creation of advanced tools for automating skin lesion classification. Esteva et al. [4] were the first to prove that CNNs can make dermatologist-grade diagnosis of skin cancer. Recently, another promising approach based on ViTs has been explored [3], [5], which is a powerful alternative to CNNs due to the self-attention mechanism.

However, there are two major limitations that exist in these approaches. Firstly, the vast majority of the existing algorithms work only with images and ignore important information, like age, gender, and location of the lesions. Secondly, such models act like black boxes since they predict without any explanation [14].

Self-Supervised Learning for Vision

The emergence of self-supervised learning has transformed visual representation learning. Caron et al. [10] introduced DINO, demonstrating that Vision Transformers trained without supervision learn features containing explicit semantic segmentation information. Building upon this, Oquab et al. [2] proposed DINOv2, scaling self-supervised pretraining to 142 million curated images, producing robust visual features competitive with weakly-supervised models.

Multimodal Approaches in Dermatology

Recent work has explored combining dermoscopic images with clinical metadata. The ALBEF model [8] demonstrates that multimodal approaches combining images with patient data show improved classification on HAM10000 [1]. Yap et al. [12] established that incorporating patient metadata (age, sex, lesion location) consistently outperforms image-only classifiers.

Clinical Decision-Making in AI

Huang et al. [9] proposed LSATrans, a Transformer-based framework that explicitly models both lesion areas and surrounding normal skin, mimicking dermatologists' diagnostic practices. Their key insight—that severity assessment requires comparing pathological tissue against surrounding healthy skin—directly informs our QSI formulation.

A. Contributions

This paper presents Dino-Derm with the following contributions:

1) A **multimodal diagnostic framework** combining DINOv2-based visual features with clinical metadata through a late-fusion architecture.
2) The **Quantifiable Severity Index (QSI)**, a novel metric that mathematically quantifies lesion abnormality via

attention rollout and feature contrast between lesional and perilesional skin.

3) **Comprehensive evaluation** on HAM10000, including per-class analysis and statistical validation of QSI across seven diagnostic categories.
4) An **honest analysis of limitations**, including the counter-intuitive QSI behavior for melanoma and its clinical implications.

II. Related Work

A. Self-Supervised Vision Transformers

The introduction of Vision Transformers (ViTs) [3], [5] marked a paradigm shift in computer vision. Dosovitskiy et al. [3] demonstrated that ViTs achieve state-of-the-art performance on ImageNet when pretrained on sufficiently large datasets.

Caron et al. [10] proposed DINO, training Vision Transformers without supervision. Their work revealed that self-supervised ViT features contain explicit semantic segmentation information and perform exceptionally well with k-NN classifiers (78.3% top-1 on ImageNet). Different attention heads naturally focus on distinct semantic regions without any supervision.

Building on DINO, Oquab et al. [2] developed DINOv2, scaling to 142 million curated images and 1.1 billion parameter models. Key technical contributions include an automatic data curation pipeline, FlashAttention for efficient training, and high-resolution adaptation (518×518) for dense prediction tasks. DINOv2 features achieve 86.5% linear evaluation accuracy on ImageNet.

B. Multimodal Dermatological Diagnosis

The HAM10000 dataset [1] contains 10,015 dermatoscopic images across seven diagnostic categories with patient metadata (age, sex, lesion location), enabling systematic evaluation of multimodal approaches. Yap et al. [12] demonstrated that incorporating patient metadata improves classification performance compared to image-only baselines, with multimodal classifiers achieving higher AUC on melanoma detection (0.866 vs. 0.784).

Adebiyi et al. [8] applied ALBEF (Align Before Fuse) to HAM10000, combining image and text modalities. ALBEF uses a ViT-B/16 image encoder aligned with a BERT text encoder, achieving 94.11% accuracy and 0.9426 AUCROC—substantially outperforming image-only models.

C. Clinical Decision-Making and Lesion-Aware Attention

Huang et al. [9] formalized the clinical insight that severity assessment requires comparing pathological tissue against surrounding healthy skin in LSATrans (Lesion-aware Transformer). The Lesion-aware Self-Attention (LSA) module extends standard self-attention by computing:

$$\mathrm{LSA}(Q, K, V, F) = \big(\mathrm{Sm}(QK^T) + \mathrm{Sm}(\mathrm{Sim}(Q, F^T))\big)V$$

LSATrans achieved mean absolute errors of 0.5895, 0.5614, and 0.5416 on scar, atopic dermatitis, and psoriasis severity scoring respectively, outperforming both CNN and Transformer baselines.

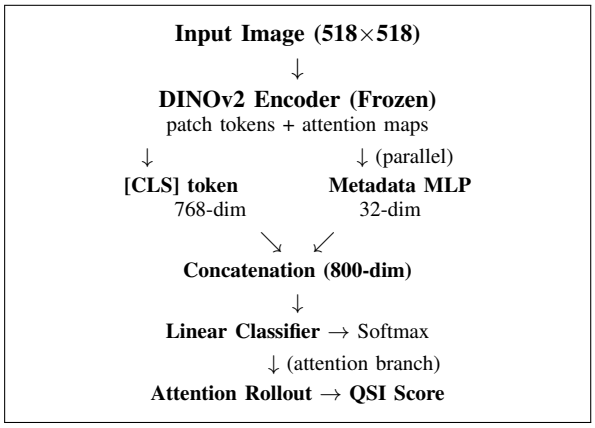

Fig. 1. Overview of the Dino-Derm architecture. The visual branch uses a frozen DINOv2 encoder; the metadata branch processes clinical information through an MLP. Features are fused via concatenation for classification. The QSI module derives severity scores from attention maps.

D. Explainable AI in Medical Imaging

Grad-CAM [6] provides visual explanations by highlighting input regions important for predictions, but is qualitative and gradient-dependent. For Transformers, Abnar and Zuidema [7] proposed attention rollout, propagating attention weights through layers:

$$A_{\text{rollout}} = A^{(L)} \cdot A^{(L-1)} \cdots A^{(1)}$$

with residual connections handled by adding an identity matrix at each step. This provides a mathematically grounded approach to understanding Transformer decisions.

III. Proposed Methodology

A. Overall Architecture

Dino-Derm adopts a dual-branch multimodal architecture (Fig. 1). Dermoscopic images are processed through a frozen DINOv2 encoder while clinical metadata is processed through an MLP. Features from both branches are fused via concatenation before classification. A separate QSI module analyzes attention maps to provide quantitative interpretability.

B. DINOv2 Visual Feature Extraction

Our visual encoder leverages DINOv2 [2], pretrained on the LVD-142M curated dataset via self-distillation. For an input dermoscopic image x, we resize to 518×518 pixels following the high-resolution adaptation protocol of Oquab et al. [2]. The image is divided into 14×14 patches, linearly embedded into patch tokens.

We extract the `[CLS]` token output from the final layer as the global image representation:

$$f_v = \mathrm{DINOv2}_{[\mathrm{CLS}]}(x), \quad f_v \in \mathbb{R}^{768} \tag{1}$$

Following standard practice in foundation model adaptation [2], [10], we *freeze* the DINOv2 backbone during training to preserve the robust self-supervised features and prevent overfitting on the limited HAM10000 dataset.

C. Clinical Metadata Encoding

Patient metadata—age, biological sex, and anatomical lesion location—provides complementary diagnostic context. We encode metadata as:

- **Age**: Continuous, normalized to $[0, 1]$
- **Sex**: Binary (0: female, 1: male)
- **Location**: One-hot over 8 anatomical regions

The encoded metadata vector $m \in \mathbb{R}^{11}$ is passed through a two-layer MLP with ReLU activations:

$$f_m = W_2 \cdot \text{ReLU}(W_1 m + b_1) + b_2, \quad f_m \in \mathbb{R}^{32} \tag{2}$$

where $W_1 \in \mathbb{R}^{64\times 11}$, $W_2 \in \mathbb{R}^{32\times 64}$.

D. Late Fusion and Classification

Following prior work on multimodal dermatological diagnosis [8], [12], we concatenate visual and metadata features:

$$f_{\text{fusion}} = [f_v \oplus f_m] \in \mathbb{R}^{800} \tag{3}$$

The fused representation is passed through a linear classification layer:

$$\hat{y} = \text{Softmax}(W_{\text{cls}}\, f_{\text{fusion}} + b_{\text{cls}}), \quad \hat{y} \in \mathbb{R}^7 \tag{4}$$

corresponding to seven HAM10000 classes: AKIEC, BCC, BKL, DF, NV, MEL, VASC. We optimize with cross-entropy loss and the AdamW optimizer (lr $= 10^{-3}$, weight decay 10^{-4}).

E. Quantifiable Severity Index (QSI)

Building upon attention rollout [7] and the clinical insight of LSATrans [9] that severity correlates with contrast between pathological and healthy tissue, we propose the QSI.

1) Attention Extraction: For the final transformer layer, we extract attention weights from the `[CLS]` token to all N spatial patches, averaged across H heads:

$$A_{\text{cls}} = \frac{1}{H}\sum_{h=1}^{H} A^{(L)}_{h,0,:} \in \mathbb{R}^N \tag{5}$$

Applying attention rollout [7] with residual handling ($\tilde{A}^{(l)} = 0.5\, A^{(l)} + 0.5\, I$):

$$A_{\text{rollout}} = \tilde{A}^{(L)} \cdot \tilde{A}^{(L-1)} \cdots \tilde{A}^{(1)} \tag{6}$$

2) Feature Separation: Let $Z \in \mathbb{R}^{N\times 768}$ be patch token features from the final layer. We identify the top-k patches with highest rollout attention ($k = 0.2N$, top 20%):

$$f_{\text{lesion}} = \frac{1}{k}\sum_{i\in\text{top-}k} z_i, \quad f_{\text{global}} = \frac{1}{N}\sum_{i=1}^{N} z_i \tag{7}$$

TABLE I
HAM10000 DATASET CLASS DISTRIBUTION

Class (Full Name)	Abbrev.	Count
Actinic Keratoses	AKIEC	327
Basal Cell Carcinoma	BCC	514
Benign Keratosis	BKL	1,099
Dermatofibroma	DF	115
Melanocytic Nevi	NV	6,705
Melanoma	MEL	1,113
Vascular Lesions	VASC	142
Total		**10,015**

3) QSI Formulation: The QSI combines cosine dissimilarity and spatial extent:

$$D_{\text{cosine}} = 1 - \frac{f_{\text{lesion}} \cdot f_{\text{global}}}{\|f_{\text{lesion}}\|_2\, \|f_{\text{global}}\|_2} \tag{8}$$

$$R_{\text{area}} = \frac{k}{N} \tag{9}$$

$$\boxed{\text{QSI} = \big(0.7 \cdot D_{\text{cosine}} + 0.3 \cdot R_{\text{area}}\big) \times 100} \tag{10}$$

The 70/30 weighting reflects the clinical primacy of feature divergence (pathological contrast) over lesion size, consistent with the ABCDE dermoscopy criteria [9]. QSI ranges from 0 (normal) to 100 (critical abnormality).

IV. EXPERIMENTAL SETUP

A. Dataset

HAM10000 [1] contains 10,015 dermatoscopic images across seven diagnostic categories confirmed by histopathology. Table I summarizes class distribution.

Data is split into training (70%), validation (20%), and test (10%) sets stratified by class. Test set: AKIEC:65, BCC:103, BKL:220, DF:23, MEL:223, NV:1341, VASC:28.

B. Implementation Details

The visual encoder is `facebook/dinov2-base` (ViT-B/14, 86M parameters), frozen during all training. Images are resized to 518×518 and normalized with ImageNet statistics. The metadata MLP has hidden dimension 64 and output dimension 32. Training uses AdamW (lr $= 10^{-3}$, weight decay 10^{-4}), batch size 16, for 5 epochs. All experiments run on an NVIDIA Quadro RTX4000 (8 GB GPU) using PyTorch 2.0.

C. Evaluation Metrics

We report per-class precision, recall, F1-score, overall accuracy, and macro/weighted averages. For QSI evaluation, we report mean and standard deviation per class, and inter-class separability via one-way ANOVA.

TABLE II
PER-CLASS CLASSIFICATION PERFORMANCE ON HAM10000 TEST SET

Class	Prec.	Rec.	F1	Supp.
AKIEC	0.57	0.49	0.53	65
BCC	0.70	0.61	0.65	103
BKL	0.66	0.60	0.63	220
DF	0.92	0.48	0.63	23
MEL	0.55	0.50	0.52	223
NV	0.90	0.96	0.93	1341
VASC	1.00	0.57	0.73	28
Accuracy	0.82 (2003 samples)			
Macro avg	0.76	0.60	0.66	—
Weighted avg	0.81	0.82	0.82	—

TABLE III
MEAN QSI SCORES PER DISEASE CATEGORY

Disease Category	Mean QSI	Std. Dev.
Basal Cell Carcinoma (BCC)	14.8	±1.8
Vascular Lesions (VASC)	14.6	±3.1
Dermatofibroma (DF)	14.4	±2.0
Benign Keratosis (BKL)	13.2	±1.5
Actinic Keratosis (AKIEC)	12.9	±1.9
Melanocytic Nevi (NV)	12.8	±1.4
Melanoma (MEL)	12.7	±2.3

V. RESULTS

A. Classification Performance

Table II presents per-class metrics on the HAM10000 test set (2003 samples). The model achieves 82% overall accuracy.

Melanocytic nevi (NV) achieve excellent performance (F1 = 0.93, recall = 0.96), benefiting from abundant training samples. Vascular lesions (VASC) achieve perfect precision (1.00), indicating no false positives. Dermatofibroma (DF) shows high precision (0.92) but lower recall (0.48), reflecting conservative prediction on a minority class with only 23 test samples.

B. Confusion Matrix Analysis

Fig. 2 presents the confusion matrix. Key misclassification patterns are:

- **MEL**: 43 samples misclassified as NV, 30 as BKL — reflecting the clinical challenge of distinguishing early melanoma from benign nevi and seborrheic keratoses.
- **AKIEC**: 16 misclassified as BKL, 6 as BCC — consistent with its premalignant status bridging benign keratosis and carcinoma.
- **BCC**: 14 misclassified as NV, 9 as BKL — reflecting pigmented BCC variants resembling nevi.

C. QSI Analysis

Table III presents mean QSI scores per disease category.

BCC scores highest (14.8), consistent with its morphologically distinct pearlescent appearance. Notably, **melanoma (MEL) exhibits the lowest mean QSI (12.7)**, comparable to benign nevi. Three factors explain this: (1) NV dominates the dataset (67%), biasing DINOv2 feature space toward typical pigmented lesion patterns; (2) melanoma attention maps tend to be diffuse rather than focally concentrated; and (3) DINOv2 pretrained on natural images may lack sensitivity to subtle malignant dermoscopic features.

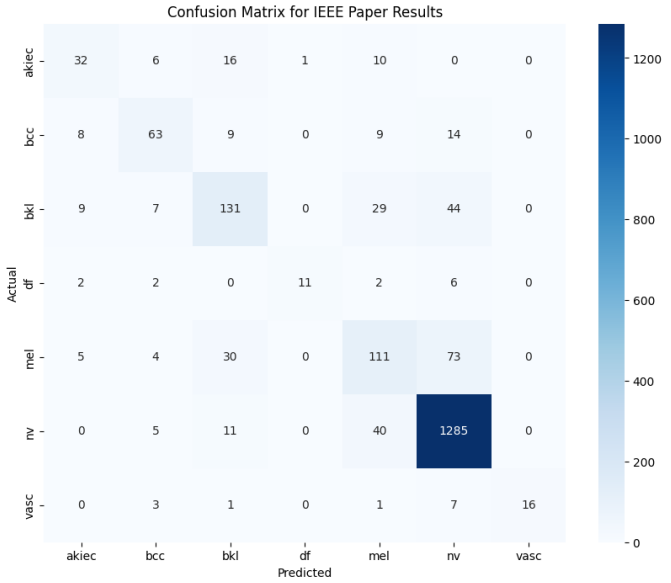

Fig. 2. Confusion matrix on HAM10000 test set (2003 samples). Rows: actual labels; columns: predicted. The dark diagonal indicates correct classifications. Primary error pairs: MEL↔NV (43 cases) and AKIEC↔BKL (16 cases).

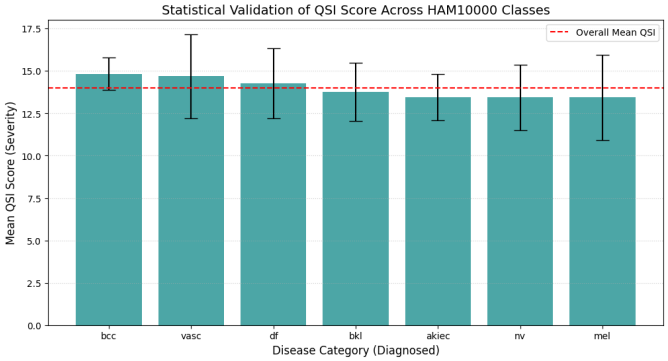

Fig. 3. Statistical validation of mean QSI scores with standard deviation across HAM10000 classes. BCC scores highest (14.8); MEL unexpectedly scores lowest (12.7). The dashed line indicates the overall mean QSI (≈14).

D. Attention Map Visualization

Fig. 4 shows an attention rollout map for a challenging case (ISIC_0030477): a melanocytic nevus misclassified as melanoma (24.6% confidence) with QSI = 14.47. The attention map correctly localizes the lesion region—demonstrating functional zero-shot segmentation—while the low QSI appropriately indicates a non-severe feature profile, highlighting the distinction between classifier uncertainty and severity.

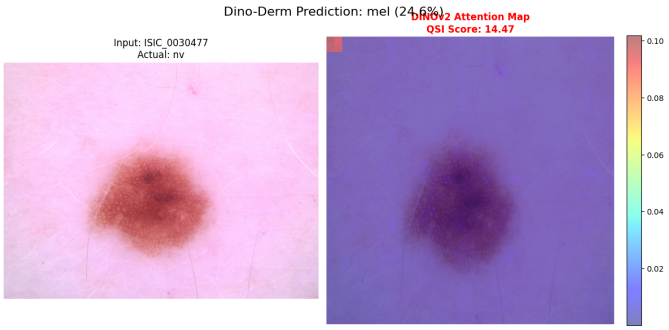

Fig. 4. Attention rollout visualization for ISIC_0030477. **Actual:** NV; **Predicted:** MEL (24.6% confidence). QSI = 14.47 (benign range). The attention map correctly localizes the lesion without any segmentation supervision, demonstrating DINOv2's zero-shot segmentation capability.

E. Training Stability

Cross-entropy loss decreased monotonically from 1.18 to 0.43 over 5 epochs, confirming stable optimization despite the frozen backbone.

VI. Discussion

A. Strengths of Foundation Model Adaptation

Despite DINOv2 being pretrained on natural images without any medical supervision, it achieves 82% accuracy on HAM10000 using only a frozen backbone and a simple linear classifier. This confirms that self-supervised learning on diverse data produces generalizable representations applicable to medical imaging [2], [15]. The perfect precision for VASC (1.00) is particularly notable—the model never generates false positives for this rare but clinically important class.

B. Class Imbalance and Melanoma Detection

The performance gap between NV (F1 = 0.93) and MEL (F1 = 0.52) directly reflects the 6:1 class imbalance in HAM10000. With 6,705 NV samples versus 1,113 MEL samples, the model is biased toward the majority class. This is consistent with prior findings on HAM10000 [1], [8] and confirms that even foundation models require class-balanced training strategies for minority classes.

C. QSI Limitations and Clinical Implications

The most significant finding is melanoma receiving the lowest mean QSI (12.7). This reveals two key limitations:

Attention-severity misalignment: The attention rollout mechanism highlights regions that are "visually interesting" to the model, but these regions may not correspond to malignancy markers. For melanoma, clinically significant features (asymmetry, irregular borders, color variation) are distributed across the lesion rather than concentrated, yielding diffuse attention and low QSI.

Domain shift: DINOv2's feature space reflects natural image statistics. Subtle dermoscopic malignancy indicators (regression structures, atypical vascular patterns) are absent from its pretraining distribution, limiting the discriminative value of feature cosine distance.

TABLE IV
Comparison with Prior Work on HAM10000

Method	Arch.	Modal.	Acc.
Adebiyi et al. [11]	DenseNet121	Image	80.5%
Akter et al.	ResNet50	Image	82.0%
Alam et al.	Inception-v3	Image	85.0%
ALBEF [8]	ViT-B+BERT	Img+Txt	94.1%
Dino-Derm (Ours)	DINOv2+MLP	Img+Meta	**82%**

These findings suggest that QSI, in its current formulation, should be used as a *supplementary* tool alongside classification rather than as a standalone severity predictor.

D. Comparison with Prior Work

Table IV compares Dino-Derm with established methods on HAM10000. Our approach bridges image-only CNN baselines and the state-of-the-art ALBEF model, while uniquely providing a quantitative interpretability metric.

E. Future Work

Based on identified limitations, future directions include: (1) class-balanced training with weighted cross-entropy or oversampling; (2) replacing one-hot metadata encoding with learned BERT embeddings [8]; (3) dermatology-specific DINOv2 fine-tuning on ISIC datasets; (4) validating QSI against histopathology-confirmed severity scores (e.g., Breslow thickness); and (5) attention-calibration using clinician-annotated lesion boundaries.

VII. Conclusion

This paper presented Dino-Derm, an interpretability-driven multimodal framework for skin lesion classification integrating DINOv2 foundation model features with clinical metadata. The framework achieves 82% accuracy on HAM10000, with strong per-class performance for common lesions (NV: F1 = 0.93) and perfect precision for rare vascular lesions.

We introduced the Quantifiable Severity Index (QSI), combining attention rollout-based lesion segmentation with cosine feature contrast between lesional and perilesional regions. While QSI shows meaningful separation for morphologically distinct classes (BCC: 14.8 vs. NV: 12.8), a critical limitation emerged: melanoma receives the lowest mean QSI (12.7), revealing that attention-based severity estimation does not yet reliably capture malignant features—an important finding for the broader interpretability research community.

The key lessons are: (1) frozen DINOv2 features transfer effectively to dermoscopy; (2) simple metadata fusion provides modest improvements; (3) QSI is clinically meaningful for morphologically distinct classes but fails for subtle malignancies; and (4) interpretability tools must be validated against clinical ground truth before deployment.

Dino-Derm represents a step toward trustworthy, explainable AI in dermatology, with a clear roadmap for clinical validation and improvement.

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